A DISCRETE DYNAMIC MODEL OF A BANK`S LIQUIDITY WITH ENDOGENOUS DEPOSITS

Ph.D. Ihor Volosyn*,
Ph.D. Anatoly Voronin**

27 December 2020

* head of the credit modeling unit of JSC “Oschadbank”,
Ihor.Voloshyn12@gmail.com

** associate professor of the department of higher mathematics and economic-mathematical methods of Simon Kuznets Kharkiv National University of Economics,
Voronin61@ukr.net

Electronic copy available at: https://ssrn.com/abstract=4319975

Abstract. The article examines the impact of endogenous deposits created by loans on the liquidity of a commercial bank. A developed discrete dynamic model of liquidity is based in particular on the application of a non-traditional indicator of turnover of liquid assets, namely, the ratio of debit turnover on loan accounts to the stock of liquid assets. The analytical decision is received using the inverse Z-transformation, and its stability is proved. The conditions for the existence of the bank's aperiodic and periodic liquidity regimes were established.

Keywords: liquidity, bank, discrete model, dynamic model, Z-transformation, liquid assets, loans, endogenous deposits, aperiodic regime, periodic regime, turnover period

JEL Classifications: G21, G28, G32

Electronic copy available at: https://ssrn.com/abstract=4319975

Introduction

Central banks have recently begun to recognize that along with Central Banks, new money (deposits) are also created by commercial banks through the granting of loans [2, 3, 4, 10, 12, 13, 14]. The Bank of England explains " Whenever a bank makes a loan, it simultaneously creates a matching deposit in the borrower's bank account, thereby creating new money. … Just as taking out a new loan creates money, the repayment of bank loans destroys money." [14].

The new deposits formed in this way do not change either the reserve of a commercial bank in its account with the Central Bank or in its cash [4]. "Although money created when a loan is granted is not Central Bank money, private customers' sight deposits at commercial banks have equal status in payment transactions …." with exogenous deposits [10].

The National Bank of Ukraine also adhered to the concept of endogenous money: "In reality, most of the money (deposits) in the modern economy is created by commercial banks through the granting of loans. Thus, when a commercial bank issues a loan, it does so by increasing the amount of the deposit on the borrower's account" [13].

Such an understanding of the formation of money (deposits) went a long and difficult way and was recognized thanks to the efforts of such researchers and practitioners as R. Ray, H. Minsky, R. Werner, W. Godley, M. Lavoie, P. Howells, B. Moore [19, 18, 6, 9, 16] and many others. R. Werner was the first to empirically prove the validity of this concept [18]. Moreover, it turned out that most of the money is created by commercial banks, not Central Banks (see, for example, [3]).

The consequence of recognizing the concept of endogenous deposits is the need to change approaches to modeling the liquidity of lending banks.

After all, those deposits expose commercial banks to the risk of withdrawing money from their accounts with the Central Bank.

Traditional models of bank liquidity do not take into account the endogenous formation of deposits during lending (see, for example, [5, 8]). In such models, the volumes of loans and deposits are independent, not related to each other. There is a limited number of works in which endogeneity is taken into account, but they investigated other issues, for example, the behavior of reserves [1], and the profit paradox [11].

The purpose of this article is to develop a liquidity model of a lending bank and to investigate the impact of endogenous deposits, i.e., created by loans, on bank liquidity.

Liquidity model

Consider a bank that has no capital and receives no profit from its operations. It is not difficult to take into account both capital and profit, but they will complicate the model and overshadow the research process. During the analysis of the bank's liquidity, we will not consider the possibility of liquidity support of it by the Central Bank as a lender of last resort.

Electronic copy available at: https://ssrn.com/abstract=4319975

Let at the beginning (t = 0) the bank has only exogenous (for instance, arrived from other banks) long-term deposits in the amount of D_e, which formed the initial stock of highly liquid assets LA_0, for example, money in a correspondent account with the Central Bank. Assume that in subsequent periods the bank does not attract new exogenous deposits and does not return existing deposits D_e. The bank uses this reserve of liquidity exclusively for making settlements between its clients and clients of other banks. Thus, the clients of this bank do not make payments among themselves within it. This is the worst-case scenario for the bank's liquidity.

The bank lends exclusively to its customers in the amount of L_t. It credits loan money to borrowers' current accounts, thereby creating endogenous demand deposits D_t. Thus, we will not consider another mechanism for issuing a loan, when the bank sends the loan money directly to another bank without credit to the borrower's current accounts.

Next, we need a settlement model between clients of this and other banks. For this purpose, for the sake of clarity, we will use the cash flow model of productive lending for the production of goods. Participants in this circulation are producers, owners of production factors, and consumers.

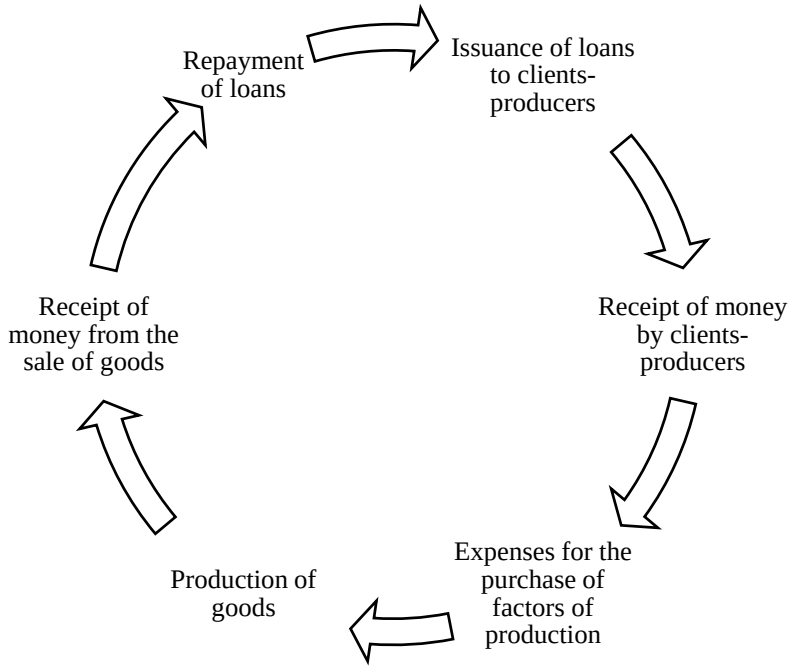


Fig. 1. Cycle of productive lending for the production of goods
compiled according to A. Graziani [6]

Without loss of generality, suppose that all producers are served in one bank, and owners of production factors and consumers do in other banks. We will assume that all participants in monetary circulation strictly adhere to the agreements concluded between them. Therefore, none of the banks is exposed to credit risk. All clients fulfill their agreements on time and in full.

For the sake of simplicity, let's assume that the bank's customers do not receive a profit from the sale, that is, they have a zero price markup. Then the money spent on

Electronic copy available at: https://ssrn.com/abstract=4319975

production is fully returned at the expense of money received from the sale of all goods. At the same time, full consumption of the manufactured goods is assumed.

According to A. Graziani [6], clients of this bank receive loans to start the production of goods. Next, clients-producers send the loan money from their current accounts (endogenous deposits) to their counterparties (owners of factors of production) to other banks for settlement of factors of production.

After the production is completed, the clients-producers receive money in their current accounts from consumers of other banks for the consumed goods. With this money in current accounts, clients-producers of this bank repay their loans in full. This ends the cycle of money circulation and begins a new one, etc.

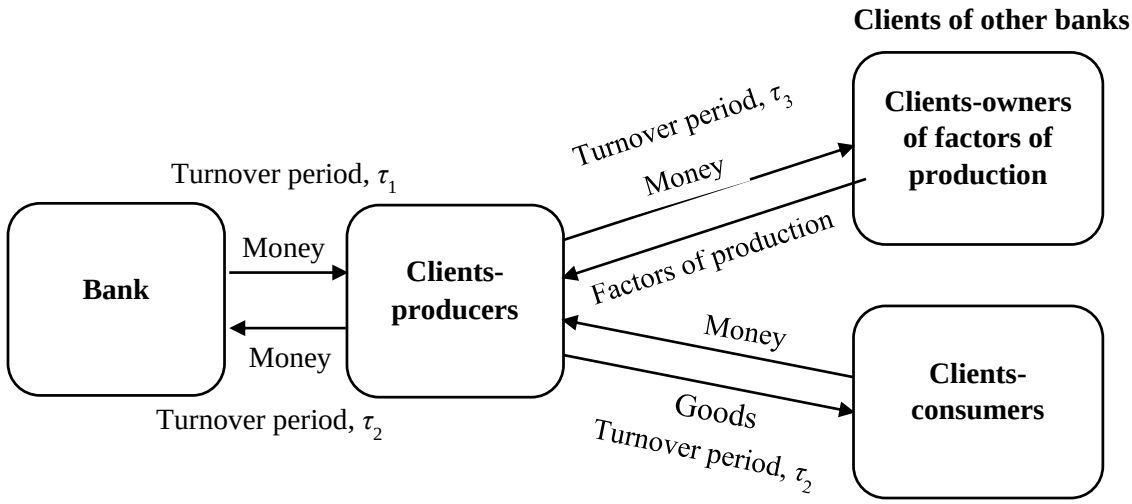


Fig. 2. Exchange of money, factors of production, and goods

Developed by the authors

Receipt and sending of money, and issuance of loans have different cycles, which are characterized by their turnover periods:

τ_1 – the turnover period of liquid assets characterizes the frequency with which the bank issues loans;

τ_2 – loan turnover period characterizes the average term for which loans are granted;

τ_3 – the turnover period of demand deposits characterizes the frequency with which the clients use money for their production needs.

When building the model, we will use general for dynamic models of the dependence between stocks and flows through the turnover period, see for example [11]:

Flow = Stock / Turnover period.

Then, the issuance of loans (debit turnover on accounts for accounting loans) obeys the following equation:

$$Dt_L = \frac{LA_t}{\tau_1},$$

loan repayment (credit turnover on accounts for accounting loans):

$$Ct_L = \frac{L_t}{\tau_2},$$

Electronic copy available at: https://ssrn.com/abstract=4319975

the outflow of deposits (debit turnover on the current account) for settlements for factors of production:

$$Dt_D = \frac{D_t}{\tau_3}.$$

The balance equation of such a bank is as follows:

$$LA_t + L_t = D_e + D_t. \quad (1)$$

Taking into account that the change in stocks is equal to the difference of receipts (inflows) and outflows and the balance equation (1), we obtain a system of non-homogeneous linear discrete equations of the first order for two variables – liquid assets LA_t and loans L_t:

$$\left.\begin{aligned} LA(t+\Delta t) - LA(t) &= -\frac{\Delta t}{\tau_3} LA(t) + \left(\frac{\Delta t}{\tau_2} - \frac{\Delta t}{\tau_3}\right) L(t) + \frac{\Delta t}{\tau_3} D_e, \\ L(t+\Delta t) - L(t) &= \frac{\Delta t}{\tau_1} LA(t) - \frac{\Delta t}{\tau_2} L(t), \end{aligned}\right\} \quad (2)$$

under initial conditions: $LA_0 = D_e$, $L_0 = 0$, $D_0 = 0$, D_e = const.

For the convenience of further explanations, we introduce dimensionless time:

$$n = t/\Delta t.$$

Let's mark $LA_t = x_1(n)$ та $L_t = x_2(n)$, $a_1 = \Delta t/\tau_1$, $a_2 = \Delta t/\tau_2$, $a_3 = \Delta t/\tau_3$.

Then the system of equations (2) will be rewritten as follows:

$$\left.\begin{aligned} x_1(n+1) &= (1- a_3)\, x_1(n) + (a_2- a_3)\, x_2(n) + a_3 D_e, \\ x_2(n+1) &= a_1\, x_1(n) + (1- a_2)\, x_2(n). \end{aligned}\right\} \quad (3)$$

The system of equations (3) makes it possible to model the utilization of the bank's liquidity reserve for lending to customers. A bank will be liquid if the stock of liquid assets remains positive throughout the entire cash flow cycle. In order to successfully solve the problems of liquidity management, it is important to determine whether the solution of system (3) will be stable, what dynamic transition regimes are possible, and what are conditions for their occurrence.

To give answers to these questions, we will apply to the system of recurrent equations (3) a discrete analog of the Laplace transform – the Z-transform:

$$\left.\begin{aligned} (z\text{-}1+ a_3)\, x_1(z) + (a_3- a_2)\, x_2(z) &= z\, x_1(0) + a_3 D_e z/(1\text{-}z) \\ - a_1\, x_1(z) + (z\text{-}1+ a_2) & \\ x_2(z) &= z\, x_2(0) \end{aligned}\right\} \quad (4)$$

The system of two algebraic equations (4) has the following solution:

$$\left.\begin{aligned} x_1(z) &= z\,\{(z\text{-}1+ a_2)\, x_1(0) + (a_2- a_3)\, x_2(0) + (z\text{-}1+ a_2)\, a_3 D_e/(z\text{-}1)\}/Q(z) \\ x_1(z) &= z\,\{a_1\, x_1(0) + (z\text{-}1+ a_3)\, x_2(0) + a_1\, a_3 D_e/(z\text{-}1)\}/Q(z) \end{aligned}\right\} \quad (5)$$

where

$$Q(z) = z^2 + b_1 z + b_2, \quad (6)$$

and

$$b_1 = a_2+ a_3\text{-}2,\ b_2 = 1\text{-}a_2- a_3+ a_2 a_3 + a_1 (a_3- a_2). \quad (7)$$

System (5) has an equilibrium position, which is determined by a system of algebraic equations:

$$\left.\begin{aligned} a_3\, x_1 + (a_3- a_2)\, x_2 &= a_3 D_e, \\ a_1\, x_1 - a_2\, x_2 &= 0 \end{aligned}\right\} \quad (8)$$

Then, the system of equations (3, 5) has the following equilibrium solution:

$$x_1^* = a_2 a_3 D_e / (a_2 a_3 + a_1 a_3 - a_1 a_2),\ x_2^* = a_1 a_3 D_e / (a_2 a_3 + a_1 a_3 - a_1 a_2) \quad (9)$$

Electronic copy available at: https://ssrn.com/abstract=4319975

Equilibrium, stationary values of balances are the levels of liquid assets, loans, and deposits that will be reached after a certain transition period when they stop changing. These are the bank's target indicators that it intends to achieve. But from the start of lending to the achievement of balanced, stationary indicators of the bank, it is still necessary to "survive" transitional regimes, which we will consider below.

Liquidity regimes

The bank remains liquid if its liquid assets $x_1^* > 0$. According to the rules of loan accounting, credit balances must also be positive $x_2^* > 0$. The positive balance of liquid assets and loans is fulfilled if:

$$a_2 a_3 + a_1 a_3 - a_1 a_2 > 0, \text{ which is identical } \tau_1 + \tau_2 > \tau_3. \quad (10)$$

Analysis of the stability of the solution shows that the fulfillment of this condition ensures the stability of the solution. Thus, small changes in turnover periods do not lead to significant, catastrophic changes in the volume of liquid assets, deposits, and loans, their movement trajectories are preserved.

Since in practice the average term of loans τ_2 is significantly longer than the turnover period of deposits τ_3, condition (10) is always fulfilled, which ensures the positivity of the equilibrium values of the balances of liquid assets, loans, and deposits, as well as the stability of the solution of the system of equations (3). For example, for annual loans τ_2 = 12 months, while the typical turnover period for deposits is $\tau_2 \leq 1$ month. Note that the ratio:

$$\tau_2 > \tau_3, \text{ or } a_3 > a_2,$$

can be called fundamental, which is generated and fulfilled by the very nature of lending.

Thus, when lending, there is always a liquidity gap that cannot be avoided. The golden rule of banking, according to which assets must be financed by liabilities of the same maturity as assets, cannot be implemented in practice. A lending bank is always exposed not only to credit risk but also to liquidity risk. Therefore, lending banks periodically require liquidity support from the Central Bank.

To study the possible regimes, we will use the discriminant analysis of characteristic quadratic equation (see formulas (6-7)):

$$Q(\lambda) = \lambda^2 + b_1 \lambda + b_2 = 0. \quad (11)$$

The discriminant (11) is equal to:

$$\Delta = (a_3 - a_2)(a_3 - a_2 - 4 a_1) \quad (12)$$

The results of the discriminant analysis of signs and the conditions for their existence are summarized in Table 1:

Table 1. Theoretically posibble liquidity regimes

No. regime	Discriminator	Conditions	Regime
1	$\Delta > 0$	$a_3 < a_2$ or $a_3 > a_2 + 4 a_1$	Aperiodic
2	$\Delta = 0$	$a_3 = a_2$ or $a_3 = a_2 + 4 a_1$	Aperiodic
3	$\Delta < 0$	$a_3 > a_2$ and $a_3 < a_2 + 4 a_1$	Periodic

It should be noted that in practice regime No. 2 cannot be used, since the average term of granting a loan always significantly exaggerates the turnover period of demand deposits. Thus, two regimes, aperiodic and periodic, are possible.

The results of modeling using system (3) are presented in Fig. 3-5.

Electronic copy available at: https://ssrn.com/abstract=4319975

Table 2. Practicaly possible liquidity regimes

No. regime	Discriminator	Conditions	Turnover periods	Regime
1	$\Delta > 0$	$a_3 < a_2$ *or* $a_3 > a_2 + 4\,a_1$	$\tau_1 = 7, \tau_2 = 12, \tau_3 = 1$	Aperiodic
3a	$\Delta < 0$	$a_3 > a_2$ and $a_3 < a_2 + 4\,a_1$	$\tau_1 = 3, \tau_2 = 18, \tau_3 = 2$	Periodic
3b*	$\Delta < 0$	$a_3 > a_2$ and $a_3 < a_2 + 4\,a_1$	$\tau_1 = 1, \tau_2 = 2, \tau_3 = 1$	Periodic

**Note. Turnover periods in regime 3b are uncharacteristic in practice, given to illustrate a higher frequency of oscillations.*

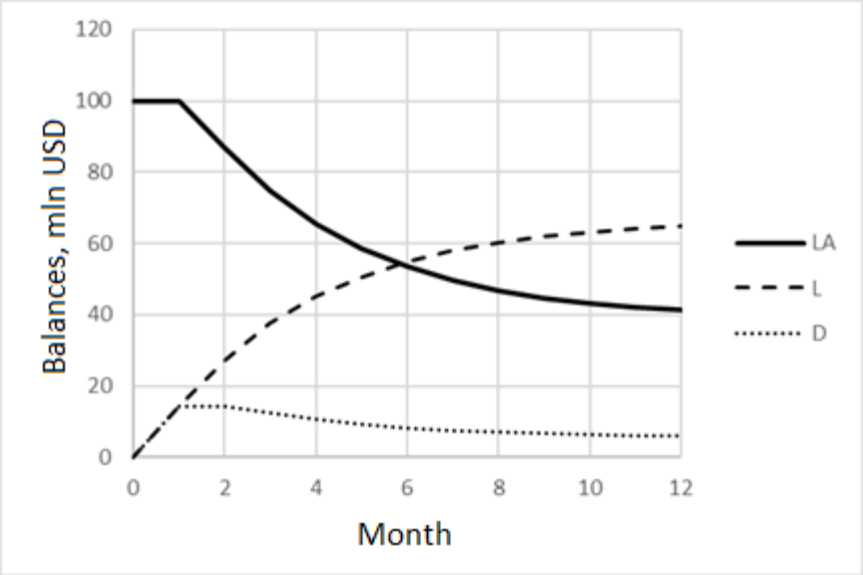


Fig. 3. Aperiodic liquidity regime ($\tau_1 = 7$, $\tau_2 = 12$, $\tau_3 = 1$)
Developed by the authors

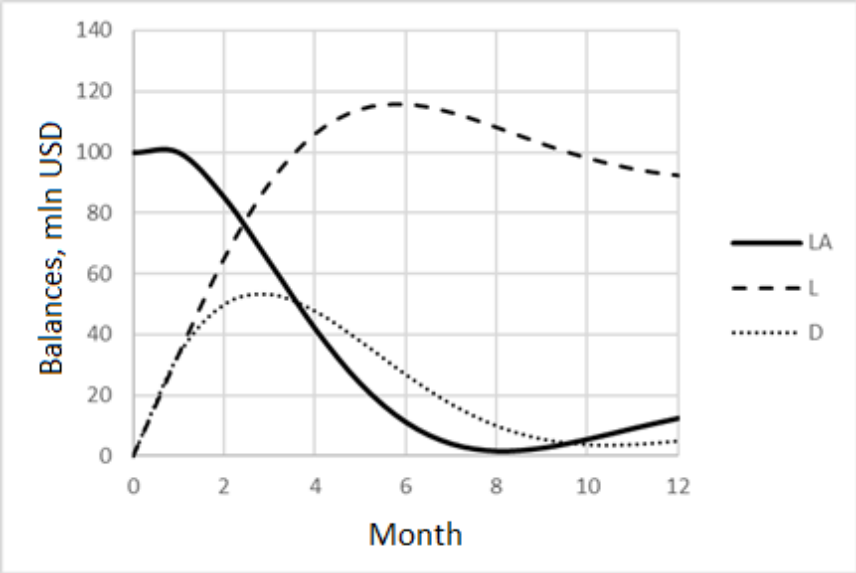


Fig. 4. Periodic regime of liquidity with a moderate frequency of fluctuations (τ_1=3, τ_2 = 18, τ_3 = 2)
Developed by the authors

Electronic copy available at: https://ssrn.com/abstract=4319975

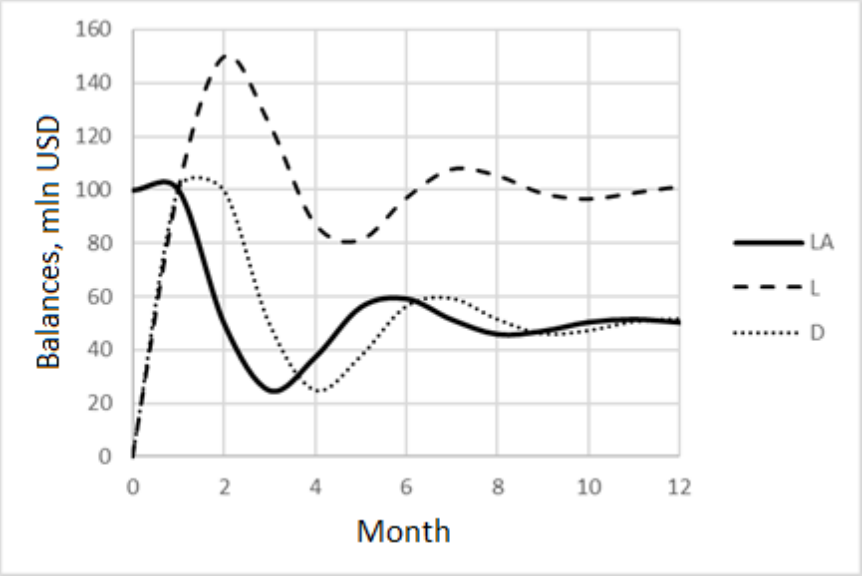


Fig. 5. Periodic regime of liquidity with a high frequency of fluctuations ($\tau_1 = 1$, $\tau_2 = 2$, $\tau_3 = 1$)

Developed by the authors

Compared to the periodic regime, the aperiodic one is more convenient and predictable for management. The bank issues loans, their volume is gradually increasing, and the liquidity reserve is gradually decreasing.

But a periodic regime can provide false signals for managerial decisions. A detailed picture of the regime (presented in Fig. 4) is shown in Fig. 6.

Starting from the 6th month, the liquidity reserve falls below the equilibrium liquidity reserve by USD 13.9 million. Liquid assets have been below the balance sheet for a long time – 8 months. The bank has to decide how it will close such a deficit of liquidity, at the expense of which money, because liquid assets are almost completely exhausted, the minimum balance is equal to UAH 1.9 million.

At the same time, the volume of loans begins to decrease after 6 months. Giving a false signal that it is necessary to change something in lending, for example, lending conditions or implementing marketing measures, etc. But the reason for such reduction in this case lies not in a change in the attractiveness of credit products for customers, or in a change in market conditions, but in the peculiarities of the behavior of a dynamic system of interconnected processes, in the presence of a periodic regime.

Note that the transition process (achieving stationary values of indicators) can last relatively long. In our example, it lasts 23 months, almost two years. Thus, in practice, bank lending is always in a transitional regime.

Electronic copy available at: https://ssrn.com/abstract=4319975

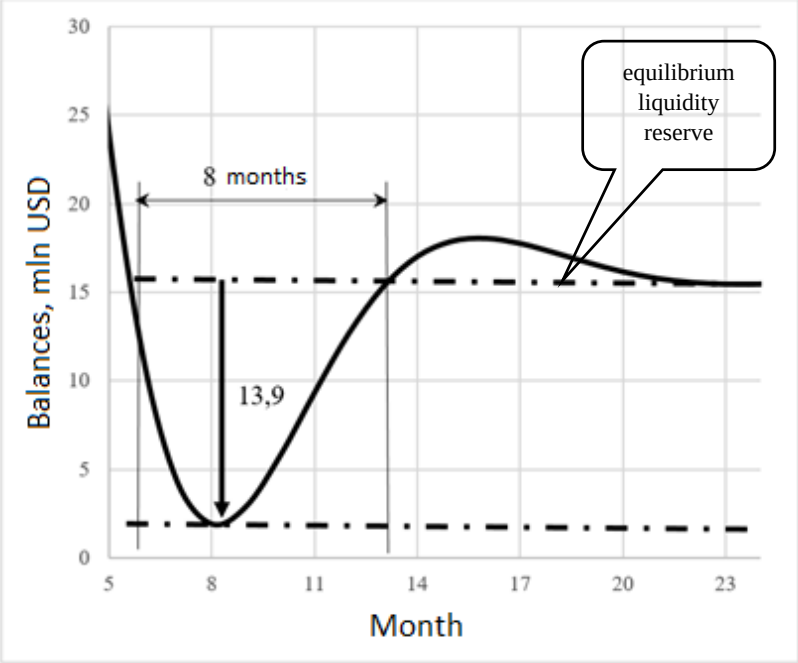


Fig. 6. A detailed picture of the reduction of the liquidity reserve below the equilibrium value

Developed by the authors

In addition, using inverse Z-transformations [15], analytical solutions of the system of equations (3) for aperiodic and periodic regimes were obtained. For the condition when the discriminant is greater than zero ($\Delta>0$), the dynamics of liquid assets in the aperiodic regime is described by the following equation:

$$x_1(\mathrm{n}) = \frac{1}{\lambda_2-\lambda_1}\left((\lambda_2+1-a_2)\lambda_2^n-(\lambda_1+1-a_2)\lambda_1^n\right)x_1(0)+ \\ +\frac{a_2-a_3}{\lambda_2-\lambda_1}(\lambda_2^n-\lambda_1^n)x_2(0)+ \\ +\left(\frac{a_2-1+\lambda_1}{(\lambda_2-\lambda_1)(1-\lambda_1)}\lambda_1^n-\frac{a_2-1+\lambda_2}{(\lambda_2-\lambda_1)(1-\lambda_2)}\lambda_2^n\right)a_3\mathrm{D}_e+x_1^*$$

and the dynamics of loans are as follows:

$$x_2(\mathrm{n}) = \frac{a_1(\lambda_2^n-\lambda_1^n)}{\lambda_2-\lambda_1}x_1(0)+\frac{(a_3-1+\lambda_2)\lambda_2^n-(a_3-1+\lambda_1)\lambda_1^n}{\lambda_2-\lambda_1}x_2(0)+ \\ +\frac{a_1a_3\mathrm{D}_e}{\lambda_2-\lambda_1}\left(\frac{\lambda_1^n}{1-\lambda_1}-\frac{\lambda_2^n}{1-\lambda_2}\right)+x_2^*,$$

where parameters $\lambda_{1,2}$ are defined as follows:

$$\lambda_{1,2}=\frac{2-a_2-a_3\pm\sqrt{\Delta}}{2},\ \lambda_2>\lambda_1.$$

For the condition when the discriminant is less than zero ($\Delta<0$), the dynamics of liquid assets in the periodic regime is described by the following equation:

$$x_1(\mathrm{n}) = \mathrm{d}^n\left(\frac{\xi d+a_2-1}{d\sqrt{1-\xi^2}}\sin(\omega n)+\cos(\omega n)\right)x_1(0) \\ +\mathrm{d}^{n-1}\frac{a_2-a_3}{\sqrt{1-\xi^2}}\sin(\omega n)\,x_2(0)+$$

Electronic copy available at: https://ssrn.com/abstract=4319975

$$+\left(\frac{\mathrm{d}^n}{a_2}\left(\frac{d(d-\xi)+(\xi d-1)(a_2-1)}{d\sqrt{1-\xi^2}}\sin(\omega n)-a_2\cos(\omega n)\right)+1\right)x_1^*,$$

and the dynamics of loans are as follows:

$$x_2(\mathrm{n}) = \mathrm{d}^{n-1}\frac{a_1}{\sqrt{1-\xi^2}}\sin(\omega n)\,x_1(0) + \mathrm{d}^n\left(\frac{\xi d+a_3-1}{d\sqrt{1-\xi^2}}\sin(\omega n)+\cos(\omega n)\right)x_2(0) + \mathrm{d}^n\left(\frac{\xi d-1}{d\sqrt{1-\xi^2}}\sin(\omega n)-\cos(\omega n)\right)x_2^*,$$

where the parameters d, ξ, and ω are defined as follows:

$\mathrm{d}^2 = (1-a_2)(1-a_2)-a_1(a_3-a_2)$, $\xi = \frac{1-a_2-a_3}{d}$ and $\omega = \arccos(\varepsilon)$.

The period of oscillations of the periodic regime is determined by the formula:

$$T = \frac{2\pi}{\omega}.$$

The dynamics of deposits can be obtained from the balance equation:

$$D(n) = x_1(n) + x_2(n) - D_e.$$

The developed discrete dynamic model makes it possible to determine:

- lending regimes (aperiodic or periodic);
- period of oscillations (fluctuations) of liquid assets;
- determine the duration and volume of the liquidity deficit relative to the equilibrium liquidity reserve;
- additionally, the time to reach equilibrium balances of liquid assets, loans, and endogenous deposits (calculation examples are not given).

For effective liquidity control, it is advisable for banks to adjust the accounting of endogenous and exogenous deposits and implement the calculation of an indicator unconventional for analysts – the turnover period of liquid assets as the ratio of debit turnover on credit accounting accounts to the stock of liquid assets.

Conclusions

The conditions for the emergence of not only the aperiodic normal regime of lending but also the little-studied periodic regime, which can significantly complicate liquidity management with false signals, are analytically established. Using the examples, the periodical regime in which there are unexpected reductions in the liquidity reserve and a drop in the volume of loans after their rapid growth is analyzed in detail. It has been established that banks actually work in transitional lending regimes, which must be detected in order to make correct management decisions. Oscillation in liquidity is explained by the nature of lending itself (determined by periods of money circulation cycles). While the change in loan balances is usually explained by changes in market conditions, demand for loans, and the macroeconomic situation. The developed model will be useful to both banking practitioners and researchers.

It is advisable to direct the following research to the development of a model that takes into account the profitability of both the bank and its customers, capital, the attraction of new exogenous deposits, and defaults of borrowers. In addition, an

Electronic copy available at: https://ssrn.com/abstract=4319975

important direction for practice is the development of a model of optimal management of the credit process.

Note that the concept of "loans generate deposits" also requires a review of approaches to prudential supervision, in particular, to assessing the liquidity and liquidity risk of commercial banks and determining the conditions under which the Central Bank should support their liquidity.

References:

1. *Blanc M., Peterson D. W.* Money Multiplier Dynamics and Banking Liquidity Cycles. https://silo.tips/download/money-multiplier-dynamics-and-banking-liquidity-cycles.
2. *Chowdhury A.K.* From Loan to Deposit: Deposit Creation by Banks and the Significance of Cash Reserves. Drishtikon: A Management Journal Volume 5 Issue 2 March 2014-September 2014. https://www.academia.edu/7840345/From_Loan_to_Deposit_Money_Creation_by_Banks_and_the_Significance_of_Cash_Reserves.
3. *Chowdhury A.K.* 'Loans First' – Explaining Money Creation by Banks. Sep 30th 2013. http://www.macroscan.org/fet/sep13/fet30092013Loans_First.htm.
4. Deutsche Bundesbank Monthly Report April 2017, Vol 69 No 4. https://www.bundesbank.de/en/publications/reports/monthly-reports/monthly-report-april-2017-667334.
5. *Fiedler R.* Liquidity Modelling. 2011 Incisive Media. https://www.academia.edu/9560965/Liquidity_Modelling.
6. *Godley W., Lavoie M.* Monetary Economics. An Integrated Approach to Credit, Money, Income, Production and Wealth.–N.Y.: PALGRAVE MACMILLAN, 2007. – 530 p.
7. *Graziani A.* The Monetary Theory of Production. Cambridge, UK: Cambridge University Press, 2003. – 176 p. https://doi.org/10.1017/CBO9780511493546.
8. *Hałaj G.* Dynamic balance sheet model with liquidity risk. Working Paper Series. No 1896 / April 2016. ECB Working Paper 1896, April 2016. https://www.ecb.europa.eu/pub/pdf/scpwps/ecbwp1896.en.pdf.
9. *Howells P.* 'The demand for endogenous money', Journal of Post Keynesian Economics, 1995, Vol. 18, No. 1, pages 89–106.
10. *Jordan T.J.* Why sovereign money would hurt Switzerland Swiss Institute of Banking and Finance at the University of St. Gallen (s/bf-HSG). Swiss National Bank, Zurich, 2018 (speech given in German). https://www.snb.ch/en/mmr/speeches/id/ref_20180503_tjn. https://www.bis.org/review/r180509a.pdf
11. *Keen S.* Solving the Paradox of Monetary Profits. Economics: The Open-Access, Open-Assessment E-Journal, Vol. 4, 2010-31. doi:10.5018/economics-ejournal.ja.2010-31. http://dx.doi.org/10.5018/economics-ejournal.ja.2010-31.
12. *Laidler D., Robson W.* 'Endogenous buffer-stock money', Credit, interest rate spreads and the monetary policy transmission mechanism, Session 3,

Electronic copy available at: https://ssrn.com/abstract=4319975

conference on The Transmission of Monetary Policy held at the Bank of Canada in November 1994.
13. Lepushinsky V., Nikolaychuk S. How money is created and what is the role of the National Bank in this process. 13624, February 26, 2019. Economy. https://voxukraine.org/uk/yak-stvoryuyutsya-groshi-i-yaka-v-tsomu-protsesi-rol-natsbanku-spojler-ne-golovna/.
14. *McLeay M., Radia A. and Thomas R.* Money creation in the modern economy. Quarterly Bulletin 2014 Q1. https://www.bankofengland.co.uk/-/media/boe/files/quarterly-bulletin/2014/money-creation-in-the-modern-economy.pdf?la=en&hash=9A8788FD44A62D8BB927123544205CE476E01654
15. Makarov I.M., Menskyi B.M. Table of inverse Laplace transforms and inverse z-transforms: Fractional-rational images.: Textbook. – Moscow: Higher School, 1978. – 247 p.
16. *Moore B.* Horizontalists and verticalists: the macroeconomics of credit money, Cambridge University Press, 1988.
17. *Wray L.R.* Modern Money Theory. A Primer on Macroeconomics for Sovereign Monetary Systems. 2-nd edition. Palgrave. Macmillan. 2012, 2015. 306p.
18. *Werner R.A.* Can banks individually create money out of nothing? — The theories and the empirical evidence. International Review of Financial Analysis. Volume 36, December 2014, Pages 1-19. https://doi.org/10.1016/j.irfa.2014.07.015. https://www.sciencedirect.com/science/article/pii/S1057521914001070.
19. *Wray L.R.* Minsky on banking: Early work on endogenous money and the prudent banker. Working Paper, January 2015, No. 827.

Electronic copy available at: https://ssrn.com/abstract=4319975